KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 3 REPORT – THE CLEANING SPRINT

Missing Value Imputation and Outlier Detection

Youth Smoking and Drug Abuse Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

Team Member Name	Roll Number
Lakshmi Prabha M M	25BAD049
Anireetha M	25BAD009
Madhumitha S	25BAD053
Atchaya S	25BAD016

1. Introduction

This report presents the Week 3 work of the project "Youth Smoking and Drug Abuse Analysis". Week 3 is the "Cleaning Sprint", in which the data-quality issues found during the Week 2 "Know Your Data" inspection are handled. The work covers missing-value imputation and outlier detection using boxplots and the Z-score method.

For this project, participants aged 18–24 are considered the youth group. This definition is used only to focus the later analysis and does not imply that the original dataset labels all participants as youth.

2. Objective

To improve the quality of the dataset by identifying and handling missing values with suitable imputation strategies, and by investigating potential outliers without removing genuine observations.

3. Activities Performed

- Created a separate copy of the original dataset (df_clean) so that the raw data remains unchanged.
- Identified the columns with missing values: Age, Gender, Education and Nscore.
- Applied median imputation to Age, Education and Nscore.
- Applied mode imputation to Gender.
- Detected potential outliers visually using boxplots.
- Detected potential outliers statistically using the Z-score method ($|Z| > 3$).
- Verified the cleaned dataset for shape, missing values and duplicates.

4. Problem Identified in Week 2

The Week 2 inspection showed that the dataset has 1,885 records and 32 columns, with no duplicate rows. However, four columns each had 113 missing values (5.99%). These gaps must be handled before exploratory analysis so that statistics and plots are not affected by incomplete records.

Column	Missing Values	Missing Percentage
Age	113	5.99%
Gender	113	5.99%
Education	113	5.99%
Nscore	113	5.99%

Table 1: Missing values before cleaning

5. Handling Missing Values

5.1 Selecting Imputation Strategies

The imputation method was chosen according to the nature of each variable. Age and Education are ordinal codes, Gender is a binary categorical variable and Nscore is a continuous personality score. The number of unique values in each column was checked first: Age has 6, Gender has 2, Education has 9 and Nscore has 49.

The median was selected for Age, Education and Nscore because it preserves the order and scale of these variables and is robust to extreme values. The mode was selected for Gender because it is the only unordered categorical variable, so a most-frequent-category replacement is the suitable choice.

Variable	Type	Unique Values	Strategy	Reason
Age	Ordinal code	6	Median	Preserves order; robust to outliers
Education	Ordinal code	9	Median	Preserves order; robust to outliers
Nscore	Continuous score	49	Median	Preserves scale; robust to outliers
Gender	Binary categorical	2	Mode	Unordered; most frequent category

Table 2: Imputation strategy for each variable

5.2 Applying Imputation

A copy of the dataset was created using df.copy(). Median imputation was applied to Age, Education and Nscore using fillna() with each column's median, and mode imputation was applied to Gender using its most frequent value.

5.3 Verifying Imputation

After imputation, the missing-value check returned an empty result, confirming that no column in the cleaned dataset contains missing values.

6. Outlier Detection

6.1 Variables Selected

Outlier analysis was applied to the seven continuous personality-score variables: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS. Boxplots and Z-scores are meant for numerical observations, so the encoded categorical columns and the CL0–CL6 drug-consumption columns were not included.

6.2 Boxplots

Boxplots were drawn to visually identify potential outliers. Points beyond the whiskers indicate values that are unusually high or low compared with the rest of the data. Nscore, Escore, Ascore and Cscore show points on both sides of the whiskers, Oscore and Impulsive show mainly high-side points, and SS shows no points beyond its whiskers.

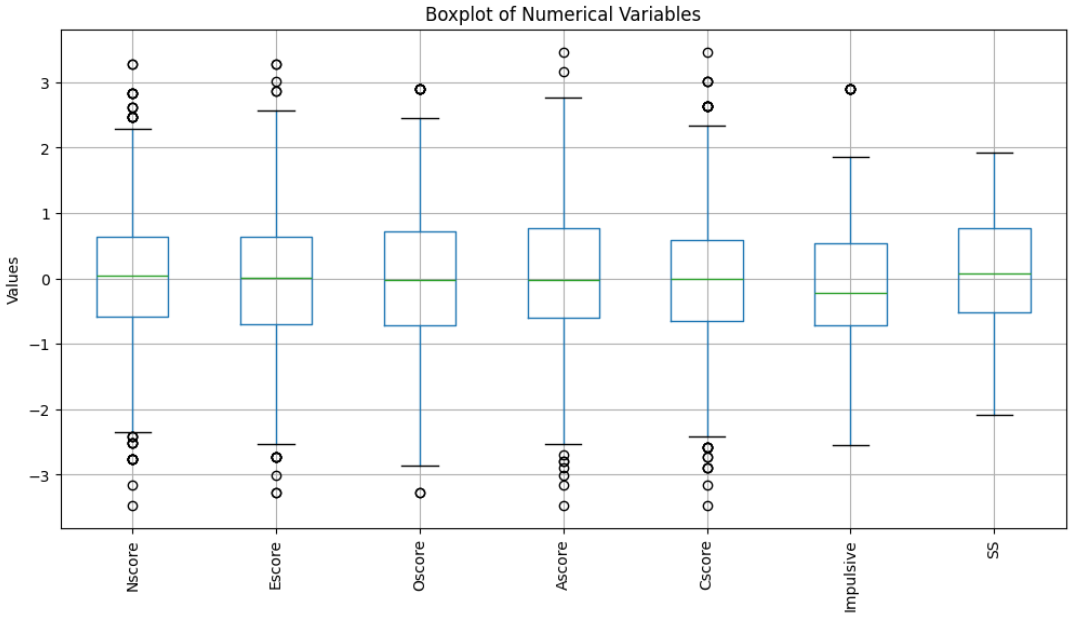


Figure 1: Boxplot of numerical variables

6.3 Z-score Method

The Z-score method identifies observations that lie far from the mean. An observation with an absolute Z-score greater than 3 was treated as a potential outlier for further investigation. The counts are shown below.

Column	Outlier Count (|Z| > 3)
Nscore	4
Escore	6
Oscore	2
Ascore	5
Cscore	6
Impulsive	7
SS	0

Table 3: Potential outliers by Z-score

Impulsive has the most potential outliers (7), followed by Escore and Cscore (6 each). SS has none. The numbers are small compared with 1,885 records, and the Z-score results agree with the points seen in the boxplots.

6.4 Decision on Outliers

The potential outliers were reported separately and were not deleted. In this dataset, extreme personality scores can be genuine observations rather than data-entry errors, and automatic removal would discard real participants.

7. Final Data Quality Check

The cleaned dataset was checked to confirm that it keeps its original size and has no missing values or duplicates.

Check	Before Cleaning	After Cleaning
Records	1,885	1,885
Columns	32	32
Missing values	452 (4 columns)	0
Duplicate rows	0	0
Potential outliers	Not checked	Identified and retained

Table 4: Data quality before and after cleaning

8. Conclusion

The dataset was cleaned by handling the missing values in Age, Gender, Education and Nscore with suitable imputation strategies. Median imputation was applied to Age, Education and Nscore, and mode imputation was applied to Gender, based on the ordinal or continuous versus categorical nature of each variable. Boxplots and the Z-score method were then used to identify potential outliers in the personality-score variables. These were reported for further investigation instead of being removed, since extreme values may be legitimate observations. The final dataset retains its original 1,885 observations and 32 variables, with no missing values and no duplicate rows.